

Research interests

2021 - Present

Graphical Models with Applications in Ecology

Application of Gaussian Graphical Models to study the structure of soil arthropod communities and to analyze the impact of various factors on belowground biodiversity in agroecosystems. The goal is to provide tools that can support decision-making in the management and protection of biodiversity.

A first methodological contribution in this area focused on the simulation-based assessment of different distance measures used to compare unweighted and undirected graphical models.

2022 - Present

Models for Ordinal Data

Extension and application of models belonging to the CUB class (Combination of discrete Uniform and shifted Binomial random variables), which allow for the measurement of two latent components: *feeling* and *uncertainty*. CUB models have wide-ranging applications, including visitor and consumer experience measurement, sports analytics, and sensory analysis. The main methodological contributions in this area include:

- the extension of the CUM model (Combination of discrete Uniform and linearly transformed Multinomial random variables) for the analysis of ordinal variables derived from semantic differential scales with five response categories;
- the definition and study of the equivalence between the basic CUB model and the CUM model;
- the development of a clustering model within the CUB framework for the analysis of multivariate rating data.

2024 - Present

Clustering and Mixture-models

An initial contribution concerned the development of a mixture model for the analysis of multivariate rating data within the CUB framework, named MLC-CUB.

I am working to relax the independence assumption in MLC-CUB through copulas, and to go over the Uniform distribution to model the uncertainty.

Education

2021 - 2024

University of Brescia — PhD in Analytics for Economics and Management

Thesis title: *Advances in Mixture Models for Ordinal Data: Theoretical Insights and Model-based Clustering*

Supervisor: Prof. Paola Zuccolotto

Co-supervisor: Prof. Julien Jacques

Final grade: Excellent (equivalent to "Ottimo" in the Italian grading system)

2018 – 2020

University of Brescia — Master's degree in Business Administration

Thesis title: *Financial Markets and Social Networks: The Effects of Covid-19*

Supervisor: Prof. Maurizio Carpita

Co-supervisor: Prof. Marica Manisera

Final grade: 110/110 with honors

2015 – 2018

University of Brescia — Bachelor's degree in Business Administration

Thesis title: *The Merge of Non-profit Organizations*

Supervisor: Prof. Isabella Maffezzoni

Final Grade: 102/110

Visiting periods

Sep. 2025 – Oct. 2025

Victoria University of Wellington — School of Mathematics and Statistics

Collaboration within the Marsden Project in developing a new model-based clustering model for mixed data with the aim of performing both classification and prediction.

Position: Visiting Researcher

Supervisors: Prof. Ivy Liu and Richard Arnold

Sep. 2023 – Jun. 2024

University Lumière of Lyon 2 — ERIC Laboratory

Development of a mixture model for the analysis of rating data within the CUB framework, named MLC-CUB

Position: Visiting PhD Student

Supervisor: Prof. Julien Jacques, director of ERIC Lab

Experience

Nov. 2024 – Oct. 2025

Postdoctoral Researcher – University of Calabria

Supervisor: Prof. Sabrina Giordano

Development and application of Copula-based Mixture Models for the analysis of multivariate rating data within the CUB framework with the aim of modeling complex dependencies within rating data.

Nov. 2024 – Oct. 2025

Postdoctoral Researcher – University of Brescia

Activity carried out within the MICS project – Supervisor: Prof. Paola Zuccolotto

Development and application of Mixture Models for the analysis of multivariate rating data and the perception, behaviors and preferences of consumers and businesses.

Mar. 2021 – Sep. 2025

Data Analyst – Siderweb S.p.A.

Activity carried out in collaboration with DMS StatLab – Director: Prof. Maurizio Carpita

Collection of data related to the steel industry and their analysis using statistical indices for the steel market and industry analysis.

Feb. 2021 – Oct. 2021

Junior Research Fellow – University of Brescia

Activity carried out in collaboration with the Agrofood Lab – Director: Prof. Gianni Gilioli

Study and application of statistical models (Regression Models, Structural Equation Models, Graphical Models, Random Forests) to investigate the impact of several factors on the edaphic biodiversity of agroecosystems, to provide useful tools to support the decisions in the management and protection of biodiversity.

Teaching Activity

Summer schools and short courses

Academic Year	Institution	Course	Duration
2024 – 2025 (7-11 July 2025)	Polytechnic University of Catalunya – Barcelona – FEM	Ordinal Data Analysis in R: Measuring Human Perceptions from Surveys	15 hours
2025 – 2026 (22 April 2026)	University of Padua – Department of Statistical Sciences	Modelling human perceptions with Mixture Models: the class of CUB models	2 hours

Teaching Assistant

Academic Year	Institution	Course	Professor in Charge	Duration
2024 - 2025	University of Brescia – DEM	Statistics	Prof. Marica Manisera	30 hours
	University of Brescia – DEM	Statistics	Prof. Ambra Macis	40 hours
2023 - 2024	University of Brescia – DEM	Statistics	Prof. Ambra Macis	40 hours
2022 - 2023	University of Brescia – DEM	Statistics	Prof. Marica Manisera	10 hours
	University of Bergamo – DISA	Statistics	Prof. Alessandro Fassò	18 hours
	University of Bergamo – DIGIP	Statistics	Prof. Alessandro Fassò	12 hours

Thesis Supervision

Co-supervisor

- | | |
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| 2024 | <ol style="list-style-type: none">1. Master's degree Thesis - <i>Statistic Analysis of Customers' Perception: Swarovski's Stores and Advertising Messages</i>2. Master's degree Thesis - <i>Statistical analysis of slow tourism: focus on the Italian trails</i> |
| 2023 | <ol style="list-style-type: none">3. Master's degree Thesis - <i>Spectral Clustering: analysis of the literature and application for the analysis of the visitor experience in an Art Gallery</i> |

Publications

Indexed Journals (WoS, Scopus)

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| 2026 | <ol style="list-style-type: none">1. Ventura, M., Macis, A., Manisera, M., & Zuccolotto, P., Measuring synesthetic experience of museum visitors using multi-point semantic differential scales. <i>Quality and Quantity</i>. Accepted for Publication.2. Ventura, M., Jacques, J., & Zuccolotto, P., Multivariate Latent Class Modeling of Rating Data for Investigating the Sustainability Perception and the Economic Behavior in the Made in Italy Sector. <i>Statistical Papers</i>. Accepted for Publication. |
| 2025 | <ol style="list-style-type: none">3. Simonetto, A., & Ventura, M. (2025). Distance measures for unweighted undirected networks: A comparison study. <i>Australian and New Zealand Journal of Statistics</i>. https://doi.org/10.1111/anzs.700154. Ventura, M., Macis, A., Manisera, M., & Zuccolotto, P. (2025). On the equivalence of two mixture models for rating data. <i>AStA Advances in Statistical Analysis</i>, 109(2), 387-411. https://doi.org/10.1007/s10182-024-00513-25. Ventura, M., Jacques, J., & Zuccolotto, P. (2025). Model-Based Clustering of Multivariate Rating Data Accounting for Feeling and Uncertainty. <i>Journal of Classification</i>, 1-22. https://doi.org/10.1007/s00357-025-09521-6 |
| 2024 | <ol style="list-style-type: none">6. Manisera, M., Migliorati, M., Ventura, M., & Zuccolotto, P. (2024). A Mixture Model for the Analysis of Categorical Variables Measured on five-point Semantic Differential Scales. <i>Austrian Journal of Statistics</i>, 53(3), 70–86. https://doi.org/10.17713/ajs.v53i3.1744 |
| 2022 | <ol style="list-style-type: none">7. Ventura, M., & Ricciardi, R. (2022). Exploring financial microblogs: Analysis of Users' Trading Profiles with Multivariate Statistical Methods. <i>Statistica e Applicazioni</i>, 20(1), 61–78. DOI: 1026350/999999_000051 |

- 2021
8. Ghiglieno, I., Simonetto, A., Sperandio, G., Ventura, M., Gatti, F., Donna, P., Tonni, M., Valenti, L., & Gilioli, G. (2021). Impact of environmental conditions and management on soil arthropods communities in vineyard ecosystems. *Sustainability*, 13(21), 11999. <https://doi.org/10.3390/su132111999>

Conference Proceedings

- 2026
1. Ventura M., Karlis D., Jacques J., Zuccolotto P., *Estimation of High-Dimensional Copula-Based CUB Models using Composite Likelihood*, [Short paper]. In K. Jajuga, P. Brito, F. Greselin, M. Zenga, & F. Denti (Eds.), *Springer Series "Studies in Classification, Data Analysis, and Knowledge Organization"*.
- 2025
2. Ventura, M., Jacques, J., & Zuccolotto, P. (2025). *Model-Based Clustering of Multivariate Rating Data to Evaluate Consumer Perceptions on Sustainability* [Short paper]. In G. Boccuzzo, E. Bovo, M. Manisera, & L. Salmaso (Eds.), *IES 2025 - Statistical Methods for Evaluation and Quality* (pp. 983-989). Cleup sc. ISBN: 9788854958494
 3. Ventura, M., Jacques, J., & Zuccolotto, P. (2025). *Clustering multivariate rating data within the CUB framework* [Short paper]. In A. Pollice & P. Mariani (Eds.), *Methodological and Applied Statistics and Demography II. SIS 2024, Short Papers, Solicited Sessions* (pp. 637–642). Italian Statistical Society Series on Advances in Statistics (ISSAS). Springer. ISBN: 9783031644467
- 2022
4. Simonetto, A., Ventura, M., & Gilioli, G. (2022). *An explorative analysis of different distance metrics to compare unweighted undirected networks* [Short paper]. In A. Balzanella, M. Bini, C. Cavicchia, & R. Verde (Eds.), *51st Scientific Meeting of the Italian Statistical Society – Caserta, 22–24 June 2022* (pp. 1522–1527). Pearson. ISBN: 9788891932310
- 2021
5. Simonetto, A., Ventura, M., Ghiglieno, I., Gatti, F., & Gilioli, G. (2021). *Le risposte della biodiversità edafica alla gestione degli agroecosistemi: Un approccio basato sull'Ecological Network Analysis* [Abstract]. In A. Elia & G. Conversa (Eds.), *Agriculture, Environment and Health. XIII National Congress on Biodiversity – Foggia, 7–9 September 2021* (p. 236). DAFNE, University of Foggia. ISBN: 9788874271016
 6. Ghiglieno, I., Simonetto, A., Sperandio, G., Ventura, M., Gatti, F., Donna, P., Tonni, M., Valenti, L., & Gilioli, G. (2021). *Impatto delle condizioni ambientali e della gestione del suolo sugli artropodi edafici dell'ecosistema vigneto* [Abstract]. In A. Elia & G. Conversa (Eds.), *Agriculture, Environment and Health. XIII National Congress on Biodiversity – Foggia, 7–9 September 2021* (p. 123). DAFNE, University of Foggia. ISBN: 9788874271016

Work in progress

1. Carloni E., Codini A. P., Ventura M., Does country of origin extend to sustainable products? The case of sustainable Made in Italy
2. Ventura M., Giordano S., Karlis D., Zuccolotto P., *Estimation of High-Dimensional Models for Ordinal Data using Pairwise Likelihood, Draft*

Conference Contributions

Invited Talks

- Speaker
1. Ventura M., Jacques J., Zuccolotto P., *Model-Based Clustering of Multivariate Rating Data to Evaluate Consumer Perceptions on Sustainability*, 12th Scientific Meeting of the Statistics for the Evaluation and Quality of Services Group of the Italian Statistical Society (IES25 - Statistical Methods for Evaluation and Quality), Brixen (Italy), 25-27 June 2025.
 2. Ventura M., Jacques J., Zuccolotto P., *Sustainability and Consumer Choice: Clustering Attitudes in the Italian Furniture Market*, StaTalk2025, Milan (Italy), 13-14 June 2025

3. Ventura M., Macis A., Manisera M., Zuccolotto P., *Synesthetic Experience in Museums: a Statistical Analysis based on Semantic Differential Scales*, 29th Nordic Conference in Mathematical Statistics (NORDSTAT), Gothenburg (Sweden), 19-22 June 2023.

Co-author

1. Macis A., Ventura M., Manisera M., Zuccolotto P., *Modeling Rating Data: Exploring the Relationship between CUB and CUM Models*, 29th Nordic Conference in Mathematical Statistics (NORDSTAT), Gothenburg (Sweden), 19-22 June 2023.

Contributed Talks

Speaker

7. Ventura M., Karlis D., Jacques J., Zuccolotto P., *Estimation of High-Dimensional Copula-Based CUB Models using Composite Likelihood*, 19th conference of the International Federation of Classification Societies, Milan (Italy), 14-16 July 2026.
8. Ventura M., Jacques J., Zuccolotto P., *Clustering Multivariate Rating Data within the CUB Framework*, 52nd Scientific Meeting of the Italian Statistical Society, Bari (Italy), 17-20 June 2024.
9. Ventura M., Macis A., Manisera M., Zuccolotto P., *A Mixture Modelling Approach to Enhance the Multisensory Experience of Museum Visitors*, Joint Conferences Data Science, Statistics and Visualisation (DSSV) and European Conference on Data Analysis (ECDA), Antwerp (Belgium), 5-7 July 2023.

Co-author

1. Macis A., Ventura M., Manisera M., Zuccolotto P., *Exploring the Equivalence of two Mixture Models for Rating Data in the CUB class*, Joint Conferences Data Science, Statistics and Visualisation (DSSV) and European Conference on Data Analysis (ECDA), Antwerp (Belgium), 5-7 July 2023.
2. Simonetto A., Ventura M., Gilioli G., *An Explorative analysis of Different Distance Metrics to Compare Unweighted Undirected Networks*, 51st Scientific Meeting of the Italian Statistical Society, Caserta (Italy), 22-24 June 2022.
3. Simonetto A., Ventura M., Ghiglieno I., Gatti F., Gilioli G., *Soil Biodiversity Responses to Agroecosystem Management: An Ecological Network Analysis Approach*, Agriculture, Environment and Health XIII National Congress on Biodiversity, Foggia (Italy), 7-9 September 2021.
4. Ghiglieno, I., Simonetto, A., Sperandio G., Ventura M., Gatti F., Donna P., Tonni M., Valenti L., Gilioli., *Impact of Environmental Conditions and Soil Management on Soil Arthropods in Vineyard Ecosystem*, Agriculture, Environment and Health XIII National Congress on Biodiversity, Foggia (Italy), 7-9 September 2021.

Posters

1. Ventura M., Giordano S., Karlis D., Zuccolotto P., *A Pairwise Likelihood Approach for High-dimensional Models for Ordinal Data*, 40th International Workshop on Statistical Modelling (IWSM), Oslo (Norway), 28 June – 3 July 2026.
2. Ventura M., Jaccques J., Zuccolotto P., *A Mixture of Multivariate CUB Models for Clustering Rating Data*, 30th Summer Working Group on Model-Based Clustering (WGMBC), Bertinoro (Italy), 22-27 July 2024.

Workshops and Seminars

Organizer/Chair

1. Seminar “An Overview of Regression Methods with Functional Data”, Prof. Julien Jacques, University Lumière Lyon 2, May 14th, 2025. University of Brescia, Brescia (Italy)
2. Seminar “Topological Data Analysis in Natural Language Processing”, Irina Proskurina, University Lumière Lyon 2, April 9th, 2025. University of Brescia, Brescia (Italy)
3. Workshop “Statistical Methods and Models for Ordinal Data”, May 25th, 2023. University of Brescia, Brescia (Italy)

Speaker

1. Seminar “*Clustering and Dependence Modeling for Multidimensional Rating Data*”, Victoria University of Padua – Department of Statistical Sciences, Padua (Italy), April 23rd, 2026.
2. Seminar “*Clustering Rating Data within the CUB framework: a Case Study on Sustainability for Made in Italy products*”, Victoria University of Wellington – Department of Mathematics and Statistics, Wellington (New Zealand), October 20th, 2025.
3. Seminar “*Finite Mixtures of CUB Models for the Analysis of Consumer Perceptions on Sustainable Made in Italy*”, Victoria University of Auckland – Department of Statistics, Auckland (New Zealand), September 10th, 2025.
4. Seminar “*Advances in the Analysis of Human Perceptions within the CUB Framework*”, Polytechnic University of Catalonia – Department of Statistics and Operation Research, Barcelona (Spain), February 4th, 2025.
5. Workshop “*Statistical Methods and Models for Ordinal Data*”, Contribution: “*Comparing CUB and CUM models*”, University of Brescia – Department of Economics and Management, Brescia (Italy), May 25th, 2023.

Softwares

CUBClustR Developed an R library for multivariate model-based clustering of rating data available via the BODaI research group repository (University of Brescia) at the following link:
<https://bodai.unibs.it/cub/mlc-cub/>

Research Projects

Participant

2022

1. **MICS – Made in Italy Circular and Sustainable**

Founded by PNRR.
Scientific coordinator: Prof. Paola Zuccolotto
Role: Postdoctoral Researcher

Description: Development of questionnaires to analyze consumer perceptions of sustainability and the Made in Italy brand. Design and application of a clustering model for rating data.

2022

2. **DS4BS – Data Science for Brescia - Arts and Cultural Places**

Founded by Fondazione Cariplo within the grant “Data Science for Science and Society”.
Scientific coordinator: Prof. Marica Manisera
Role: Research collaboration

Description: Development and application of Mixture Models to analyze data on visitor behaviour at cultural sites in Brescia to improve their experience and attract more visitors, ultimately contributing to increased tourism and cultural activities in the region.

2021

3. **F.A.Re.Su.Bio. – Fertilità, Ambiente e Reddito attraverso Suolo e Biodiversità**

Project under the PSR (Rural Development Plan) funded by Regione Lombardia
Scientific coordinator for UniBs: Prof. Gianni Gilioli
Role: Research collaboration

Description: Application of machine learning models (random forests, CART, GLM), Structural Equation Models (SEM) and Graphical Models to investigate the relationships between soil biodiversity indicators (e.g., soil biological quality) and meteorological variables, vineyard management practices, and soil chemical-physical characteristics, providing decision support tools for biodiversity management in the wine sector.

4. **Development of a sampling protocol for *Anopheles spp.* and assessment of their potential role as vectors of emerging pathogens**

Project funded by Ministry of Health
Scientific coordinator for UniBs: Prof. Gianni Gilioli

Role: Research collaboration

Description: Application of machine learning models to assess the impact of meteorological and land use factors on predicting Anopheles mosquito population abundance and to evaluate the risk of their presence using landscape and meteorological data, with the goal of developing optimal sampling protocols.

Scientific Cooperation

- Member of the Italian Statistical Society (SIS)
- Member of the Young group of the Italian Statistical Society (y-SIS)

Awards and Honors

Feb. 2026 **Marie Skłodowska Curie Action Seal of Excellence – European Commission**
Project: ACHIEVE – Advanced Clustering for Human perceptions in IntEgrated Views of wEllbeing.
Score: **91.60**/100
Description: Awarded the Seal of Excellence for a high-quality project proposal in a highly competitive evaluation. The proposal was rated "Excellent" in Research Quality, Training, and Host Institution integration (University of Brescia, Polytechnic University of Catalonia, and Victoria University of Wellington).

Academic service

Reviewer for the following journals:

- Electronic Journal of Applied Statistical Analysis (EJASA)
- Advances in Statistical Analysis (AStA)
- Statistical Papers (StPa)
- Journal of Classification (JoC)

Reviewer for PhD thesis:

2025 1. PhD thesis entitled “*Partial Ordered Stereotype Model Development of a New Model for Ordinal Data*” – Victoria University of Wellington, New Zealand – Role: Overseas examiner

Digital Skills

Microsoft | MS Office | R | Python | Latex | Lime Survey

Languages

Italian	Native				
English	Comprehension		Speaking		Writing
	Listening	Reading	Spoken production	Spoken interaction	
	French	B2	B2	B2	B2
	A2	A2	A2	A2	A2

I hereby declare that the information reported in this CV is true and complete to the best of my knowledge.

Brescia, April 8th, 2026

A handwritten signature in black ink, appearing to read 'M. Vesentini', written in a cursive style.